



In this chapter you will learn about...

- Algebraic expressions
- Types of algebraic expressions
- Degree of a polynomial
- Types of polynomials
- Finding the value of an algebraic expression
- Addition and subtraction of algebraic expressions

In class VI, we have learnt about simple algebraic expressions like $x + 4$, $y - 3$, $2x + 3$, $2y - 7$ and so on.

A symbol having a fixed numerical value is called a **constant** and a symbol which can take various numerical values is called a **variable**. In this chapter, we shall study algebraic expressions involving one or two variables. We shall also study about their addition and subtraction.

ALGEBRAIC EXPRESSION

A combination of constants and variables connected by some or all of the four fundamental operations- addition, subtraction, multiplication and division is called an **algebraic expression**.

Examples : (i) $2x^2y - 5x + 6$
 (ii) $2xy - 8$
 (iii) $y^2z + yz + 7$

Terms

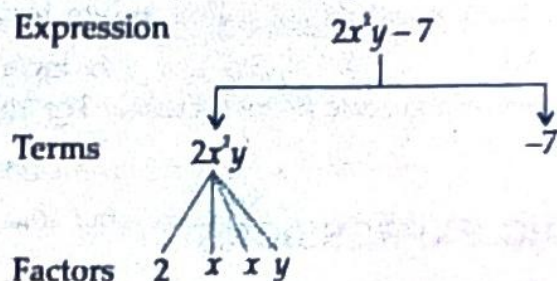
The different parts of an algebraic expression separated by the sign + or - are called the **terms** of an expression.

Examples : (i) $4 - 5x + 2x^2y$ is an algebraic expression having three terms, namely 4, $- 5x$ and $2x^2y$.
 (ii) $2x^3 - 5x$ is an algebraic expression having $2x^3$ and $- 5x$ as its terms.

Factors of a Term

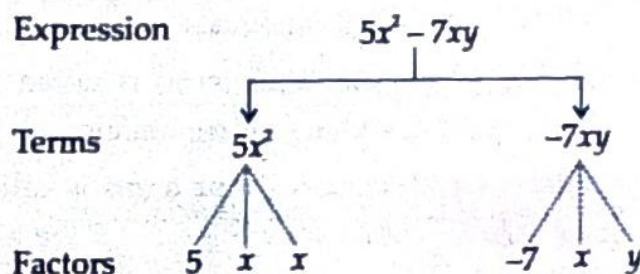
The algebraic expression $5x^2 - 2xy$ consists of two terms $5x^2$ and $-2xy$. The term $5x^2$ is a product of 5, x and x . We say that 5, x and x are the factors of the term $5x^2$. Similarly, the term $-2xy$ is a product of the factors -2 , x and y . Therefore, a term is a product of its factors.

We can also represent the terms and factors of the terms of an expression by a tree diagram. Let us draw a tree diagram for the expression $2x^2y - 7$.



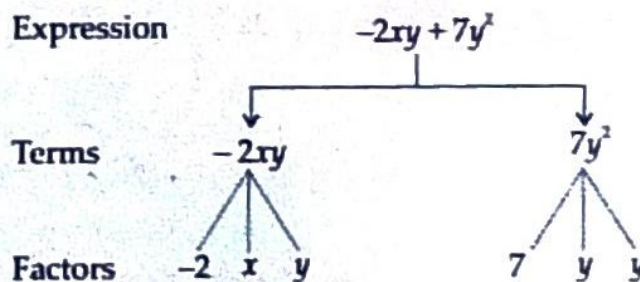
Example 1 : Represent $5x^2 - 7xy$ by a tree diagram indicating terms and factors.

Solution : Given expression is $5x^2 - 7xy$.



Example 2 : Represent the terms and factors of the terms of $-2xy + 7y^2$ by a tree diagram.

Solution : Given expression is $-2xy + 7y^2$.



Note : In the tree diagram we have used dotted lines for factors and continuous lines for terms.

Coefficients

The coefficient of a term of an algebraic expression may be numerical or variable factor or both.

Example : In the term $5x$, the numerical coefficient of x is 5 and variable coefficient of 5 is x .

When the coefficient of a term is $+1$, it is usually omitted.

Example : In $x^2y^2 + 6$, the coefficient of x^2y^2 is 1.

The coefficient -1 is indicated only by the negative sign.

Example : $-1x^2$ is written as $-x^2$.

Like and Unlike Terms

The terms having the same variable factors are called **like** terms. Like terms may differ in their numerical coefficients.

Example : In the expression $3xy - 4x + 5yx + 7$, the factors of $3xy$ are 3, x and y . The factors of $5yx$ are 5, y and x . Thus, their algebraic factors (variable) are same. Therefore, $3xy$ and $5yx$ are like terms.

The terms not having same variable factors are called **unlike** terms.

Example : In the expression $3xy - 4x + 5yx + 7$, $3xy$ and $-4x$ have different algebraic factors. Similarly, $5yx$ and 7 have different algebraic factors. Hence, $3xy$ and $-4x$ are unlike terms. $5yx$ and 7 are also unlike terms.

TYPES OF ALGEBRAIC EXPRESSIONS

1. **Monomial :** An expression which contains only **one** term is known as a monomial. Thus, x , $3x$, $-3xy$, $4x^2y$, abc , -9 are all monomials.
2. **Binomial :** An expression which contains **two** terms is called a binomial. Thus, $x - 5$, $2x + 3y$, $x^2 - 7$, $4x^2 - y$ are all binomials.
3. **Trinomial :** An expression which contains **three** terms is called a trinomial. Thus, $x^3 - y^3 + 2z^3$, $x + y - 5$, $xyz + 6 + x^2$ are all trinomials.
4. **Quadrinomial :** An expression which contains **four** terms is called a quadrinomial. Thus, $2x^2 + y^2 - z^2 + xyz$, $a^3 + 3a^2b + 3ab^2 + b^3$, $x + y + z - 6$ are all quadrinomials.
5. **Polynomial :** An expression which contains one or more terms is known as polynomial. So, monomials, binomials, trinomials and quadrinomials are all polynomials, since they have either one or more terms in all. In a polynomial, a variable cannot appear in the denominator.

Example : $3x + 5$, $7x^2 - 6x + 1$ are polynomials. $2x - \frac{3}{x} - 7$, $2x^3 + \frac{5}{x^2} - 7$ are not polynomials.

Example 1 : Write down the coefficient of :

- (i) x in $-2xy$ (ii) xyz in $-\frac{7}{2}xyz$ (iii) x in $-xyz$

Solution : (i) $-2xy = -2y(x)$

Hence, the coefficient of x in $-2xy$ is $-2y$.

(ii) $-\frac{7}{2}xyz = -\frac{7}{2}(xyz)$

Hence, the coefficient of xyz in $-\frac{7}{2}xyz$ is $-\frac{7}{2}$.

(iii) $-xyz = -yz(x)$

Hence, the coefficient of x in $-xyz$ is $-yz$.

Example 2 : Write the numerical coefficient of :

(i) xy

(ii) $-3xy$

(iii) $-9abc$

(iv) $-5xy^2z$

Solution : (i) $xy = 1(xy)$

$\therefore$ The numerical coefficient of xy is 1.

$$(ii) -3xy = -3(xy)$$

$\therefore$ The numerical coefficient of $-3xy$ is -3 .

$$(iii) -9abc = -9(abc)$$

$\therefore$ The numerical coefficient of $-9abc$ is -9 .

$$(iv) -5xy^2z = -5(xy^2z)$$

$\therefore$ The numerical coefficient of $-5xy^2z$ is -5 .

Example 3 : Write the constant term of the following polynomials :

$$(i) -2x^2 - 9 \quad (ii) 2x^2 - 5x + 6 \quad (iii) a^2 + ab - \frac{5}{2}$$

Solution : (i) The constant term in $-2x^2 - 9$ is -9 .

(ii) The constant term in $2x^2 - 5x + 6$ is 6 .

(iii) The constant term in $a^2 + ab - \frac{5}{2}$ is $-\frac{5}{2}$.

Example 4 : State whether the given pair of terms are like or unlike terms :

$$(i) 6, 99 \quad (ii) -4x, \frac{3}{2}x \quad (iii) 12ab, 12a^2b^2 \quad (iv) -7x, -7y$$

$$(v) 8xy, 12yx$$

Solution : (i) 6 and 99 are constants, hence they are like terms.

(ii) Factors of $-4x = -4, x$

$$\text{Factors of } \frac{3}{2}x = \frac{3}{2}, x$$

Since $-4x$ and $\frac{3}{2}x$ have same variable factor x , hence they are like terms.

(iii) Factors of $12ab = 12, a, b$

Factors of $12a^2b^2 = 12, a, a, b, b$

Since variable factors are different, $12ab$ and $12a^2b^2$ are unlike terms.

(iv) Factors of $-7x = -7, x$

Factors of $-7y = -7, y$

Since variable factors are different, $-7x$ and $-7y$ are unlike terms.

(v) Factors of $8xy = 8, x, y$

Factors of $12yx = 12, y, x$

Since variable factors are same, $8xy$ and $12yx$ are like terms.

Example 5 : Identify the monomials, binomials, trinomials and quadrinomials in the following :

$$(i) x^2 + y \quad (ii) 2x^2 + 4y - 6 \quad (iii) -5x^3 \quad (iv) x - 3$$

$$(v) 9x^3 - ax + c \quad (vi) 2abc \quad (vii) x^2 + y^2 + z^2 - 6 \quad (viii) 6xy$$

$$(ix) x^9 \quad (x) 3x - 1 \quad (xi) x - y - 2z \quad (xii) 7 + 4x^2y^2z$$

Solution : The monomials have only one term.

So, (iii) $-5x^3$, (vi) $2abc$, (viii) $6xy$ and (ix) x^9 are monomials.

The binomials have two terms.

So, (i) $x^2 + y$, (iv) $x - 3$, (x) $3x - 1$ and (xii) $7 + 4x^2y^2z$ are binomials.

The trinomials have three terms.

So, (ii) $2x^2 + 4y - 6$ (v) $9x^3 - ax + c$ and (xi) $x - y - 2z$ are trinomials.

The quadrinomials have four terms.

So, (vii) $x^2 + y^2 + z^2 - 6$ is a quadrinomial.

Example 6 : Write the following cases in the form of algebraic expressions using numbers, variables and signs of basic operations :

(i) The sum of number 7 and x .

(ii) One-third of x .

(iii) One-fourth of the sum of a and b .

(iv) 3 less than the difference of m and n .

(v) 2 less than the quotient of x and y .

Solution : (i) The sum of number 7 and x means 7 is added to x , i.e., $x + 7$.

(ii) One-third of x means $\frac{1}{3}$ is multiplied to x , i.e., $\frac{1}{3}x = \frac{x}{3}$

(iii) Sum of a and $b = a + b$

One-fourth of the sum of a and $b = \frac{1}{4}$ of $(a + b)$

$$= \frac{1}{4} \times (a + b) = \frac{1}{4}(a + b)$$

(iv) Difference of m and $n = m - n$.

3 less than the difference of m and $n = (m - n) - 3$

(v) The quotient of x and $y = x \div y = \frac{x}{y}$

2 less than the quotient of x and $y = \left(\frac{x}{y}\right) - 2$

Example 7 : Change the following algebraic expressions to statements in words.

(i) $3(x + y)$ (ii) $z \times x = 3$ (iii) $\frac{3}{4}(x + y)$

Solution : (i) Three times the sum of variables x and y .

(ii) Product of variables z and x is 3.

(iii) Three-fourths of the sum of variables x and y .



EXERCISE 7.1

1. Write all the terms of each of the following algebraic expressions :

(i) $5x^4 - 3x^3 + 2x + 7$

(ii) $a^2 + b^2 + 2ab$

(iii) $a^3 - ab + b^2$

(iv) $12x^3 + 5x - 8$

(v) $3x^3 - 7x^2y^2 + 7$

(vi) $4x^2y^2 - 7xy + 4$

2. Write the constant terms in the following expressions :

(i) $x^2 - 8$

(ii) $x^2 - 4x + \frac{7}{9}$

(iii) $x + \frac{1}{2}$

(iv) $3x^2 - 5x - 3$

3. Identify the monomials, binomials, trinomials and quadrinomials from the following expressions :

(i) $a^2 - b^2$

(ii) $7x^3$

(iii) $ax^2 - bx - c$

(iv) $-xyz$

(v) $7xy^2$

(vi) $2x + 3$

(vii) $x^2 + y^2 - z^2 + 1$

(viii) $6ab^3$

(ix) $2x^2 - 5x + 7$

(x) $3a - 2a$

4. Write down the coefficient of x in each of the following :

(i) $-2x$

(ii) $15xy^3$

(iii) $-\frac{7}{3}xyz^2$

(iv) $7 - \frac{3}{2}x$

(v) $-\frac{7}{2}ax$

(vi) $-\frac{5}{9}xy + 7$

(vii) $4xy^2 + 2y^2z$

(viii) $-6xy^5$

5. Change the following algebraic expressions to statements in words :

(i) $x - y$

(ii) $4(x + y)$

(iii) $7y + 5$

(iv) $2a - b$

(v) $\frac{x}{y} - 6$

(vi) $x \times x$

(vii) $3mn + 7$

(viii) $ab - (a + b)$

6. Identify the like and unlike terms in the following :

(i) $-5xy, 3xy^2, 3x^2y$

(ii) $3a, 5a, 7a, -6a$

(iii) $4x^2y, 3xy^2, 6b^2z, -8bz^2$

(iv) $5x, 7y, -3z$

(v) $9ab^2, -4ab^2, ab^2$

(vi) $4xy^2z, -6y^2xz, 7xzy^2$

7. Identify the like terms in each of the following :

(i) $3x^2, y^2, -x^2, z^2$

(ii) $2ab, bc, 6a, -4bc$

(iii) $3x^2z, -4x^2y, -6yx^2, x^2y^2$

(iv) $x^2yz, yxz^2, zx^2y, xy^2z, 6xyz^2$

8. Give the algebraic expressions in the following cases using variables, constants and arithmetic operations :

- (i) Sum of the squares of x and y . (ii) Product of x and y subtracted from 9.
 (iii) 5 less than the product of x and y^2 .
 (iv) Number 8 added to twice the product of x and y .
 (v) 6 added to the difference of x and $2y$. (vi) 7 added to x multiplied by itself.

Degree of a Polynomial

The highest power of the variable in a polynomial is called its **degree**.

If a polynomial consists of more than one variable, then the sum of the powers of variables in its different terms determine its degree. The degree of the polynomial in such case is the highest sum of the powers.

Case 1 : Polynomial with only one variable i.e., $7x^2 + 4x + 6$.

This is a polynomial in one variable i.e., x and the highest power of x is 2. Hence, the degree of the polynomial is 2.

Case 2 : Polynomial with more than one variable. i.e., $8x^2y + 3xy + 4x^3y^2$.

Here degree of term $8x^2y = 2 + 1 = 3$, degree of term $3xy = 1 + 1 = 2$ and degree of term $4x^3y^2 = 3 + 2 = 5$.

Since the highest degree is '5', hence 5 is the degree of the polynomial.

Types of Polynomials

1. Linear polynomial : A polynomial of degree '1' is called a linear polynomial.

Examples : $2x + 3$, $-5x$, $\frac{3}{2}y + 7$

2. Quadratic polynomial : A polynomial of degree '2' is called a quadratic polynomial.

Examples : $x^2 + 5x - 6$, $2x^2 - 7$, $xy + 2x - 6$

3. Cubic polynomial : A polynomial of degree 3 is called a cubic polynomial.

Examples : $x^2y + 2xy - 5$, $x^3 - 5x^2 + 6$, $6x^3 - 7$

Example 1 : Write the degree of the following polynomials :

(i) $-x^2 + 5x$

(ii) $4x^2y + 5xy - 4x$

(iii) $3x^2 - 2xy^2 - 5xy$

Solution : (i) In polynomial $-x^2 + 5x$, highest power is 2, hence, its degree is 2.

∴ A polynomial of degree '4' is called by quadratic polynomial

(ii) In polynomial $4x^2y + 5xy - 4x$, there are two variables x and y .
 degree of the term $4x^2y = 2 + 1 = 3$
 degree of the term $5xy = 1 + 1 = 2$
 degree of the term $-4x = 1$

Since highest degree is 3, hence the polynomial is of degree 3.

(iii) In polynomial $3x^2 - 2xy^2 - 5xy$, the term having highest degree is $-2xy^2$ whose degree is $1 + 2 = 3$.

Hence, polynomial is of degree 3.

Example 2 : Name the following expressions on the basis of number of terms and degree of the polynomials :

$$(i) \quad 6x^2 + 2x - 5 \qquad (ii) \quad \frac{1}{2}x - 6 \qquad (iii) \quad 4a^2b$$

Solution : (i) The algebraic expression $6x^2 + 2x - 5$ has three terms $6x^2$, $2x$ and -5 , therefore, it is a trinomial. The highest power is of the term $6x^2$ i.e., 2, hence it is quadratic polynomial.

(ii) The algebraic expression $\frac{1}{2}x - 6$, has two terms $\frac{1}{2}x$ and -6 , therefore, it is a binomial. The degree of the polynomial is 1, hence it is a linear polynomial.

(iii) The algebraic expression $4a^2b$ has only one term, therefore, it is a monomial. The degree is 3, hence it is a cubic polynomial.

FINDING THE VALUE OF AN ALGEBRAIC EXPRESSION

We know that an algebraic expression contains one or more terms. Each term further contains one or more variables and some numerical coefficients.

If the values of all literal (variables) are known in a given expression then on replacing these variables by their numerical values, we can obtain an arithmetic expression which can be evaluated. The process of replacing the variables by their numerical values is called **substitution**.

Example 1 : If $x = 1$, $y = 3$ and $z = -1$, find the value of

$$(i) \quad 5x + 3y - xyz \qquad (ii) \quad 2x^2 - 7x^3y + z^2 - 6 \qquad (iii) \quad 4x^2y + 3xz - 10$$

Solution : We have, $x = 1$, $y = 3$ and $z = -1$

$$\begin{aligned} (i) \quad 5x + 3y - xyz &= 5(1) + 3(3) - (1)(3)(-1) \\ &= 5 + 9 + 3 \\ &= 17 \end{aligned}$$

$$\begin{aligned} (ii) \quad 2x^2 - 7x^3y + z^2 - 6 &= 2(1)^2 - 7(1)^3(3) + (-1)^2 - 6 \\ &= 2 - 21 + 1 - 6 \\ &= 3 - 27 \\ &= -24 \end{aligned}$$

$$\begin{aligned} (iii) \quad 4x^2y + 3xz - 10 &= 4(1)^2(3) + 3(1)(-1) - 10 \\ &= 12 - 3 - 10 \\ &= 12 - 13 = -1 \end{aligned}$$

Example 2 : If $a = -2$, $b = -3$ and $c = -4$, evaluate the following :

$$(i) \quad a^2 + b^2 - c \qquad (ii) \quad ab - c \qquad (iii) \quad a^3 + b^3 + c^3 + 3abc$$

Solution : We have, $a = -2$, $b = -3$ and $c = -4$

$$\begin{aligned} (i) \quad a^2 + b^2 - c &= (-2)^2 + (-3)^2 - (-4) \\ &= 4 + 9 + 4 = 17 \end{aligned}$$

$$\begin{aligned} (ii) \quad ab - c &= (-2)(-3) - (-4) \\ &= (2 \times 3) + 4 \end{aligned}$$

$$= 6 + 4 = 10$$

$$\begin{aligned} \text{(iii)} \quad a^3 + b^3 + c^3 + 3abc &= (-2)^3 + (-3)^3 + (-4)^3 + 3(-2)(-3)(-4) \\ &= -8 - 27 - 64 + 3(-24) \\ &= -8 - 27 - 64 - 72 = -171 \end{aligned}$$

Example 3 : If $x = -1$, $y = 2$ and $z = -2$, evaluate

$$\text{(i)} \quad x^2 - 2y^2 + 4z^3 \quad \text{(ii)} \quad 3x^2y + 5xy^2 + 2xyz$$

Solution : We have, $x = -1$, $y = 2$ and $z = -2$

$$\begin{aligned} \text{(i)} \quad x^2 - 2y^2 + 4z^3 &= (-1)^2 - 2(2)^2 + 4(-2)^3 \\ &= 1 - 2(4) + 4(-8) \\ &= 1 - 8 - 32 \\ &= 1 - (8 + 32) \\ &= 1 - 40 = -39 \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad 3x^2y + 5xy^2 + 2xyz &= 3(-1)^2(2) + 5(-1)(2)^2 + 2(-1)(2)(-2) \\ &= 3(1)(2) + (-5)(4) + 8 \\ &= 6 - 20 + 8 \\ &= (6 + 8) - 20 \\ &= 14 - 20 = -6 \end{aligned}$$



Laughter Time

Q. What do you call a bird that doesn't eat?

A. Polynomial (Poly no meal).



Full to do Examples **EXERCISE 7.2**

1. Write the degree of the following polynomials:

$$\text{(i)} \quad 6xy + 9xy^2 - 2y^3 + x^4$$

$$\text{(ii)} \quad 2a^2 - b^2 + 7a^2b - 1$$

$$\text{(iii)} \quad 7x^2 + 8y^2 - 6x + \frac{1}{3}y$$

$$\text{(iv)} \quad \frac{2}{3}x^2 + 7x^3y + 6xy - 5$$

$$\text{(v)} \quad 3x^2 - 4y^3 + 2x^3y - 6x$$

$$\text{(vi)} \quad 15x^3 - 2xy + 7x^2$$

2. Name the following expressions on the basis of number of terms and degree of polynomials.

$$\text{(i)} \quad 7x^2$$

$$\text{(ii)} \quad a - 9$$

$$\text{(iii)} \quad \frac{1}{3}x + \frac{2}{5}y$$

$$\text{(iv)} \quad x^2 + 5x - 11$$

$$\text{(v)} \quad x^3 - 7x^2 - x$$

$$\text{(vi)} \quad -9x^2y$$

3. Classify the following as linear, quadratic and cubic polynomials :

$$\text{(i)} \quad 6a + 3b - c$$

$$\text{(ii)} \quad 4x^2 - 2xy + y^2$$

$$\text{(iii)} \quad x^3 - xy + 7$$

$$\text{(iv)} \quad x^3 + 7x^2 - 5x + 2$$

$$\text{(v)} \quad \frac{2}{3}x - 7$$

$$\text{(vi)} \quad 7xy^2$$

4. Evaluate each of the following algebraic expressions, if $x = -1$, $y = 2$, $z = -2$.

(i) $2x^2 - y^2$

(ii) $2xy^2 - 3yz + xz^2$

(iii) $\frac{x^3 + y^2 + z}{3}$

(iv) $2x + 3y - \frac{3}{2}z$

(v) $5 + 4x^3 - 2z^2$

5. Evaluate each of the following expressions given that $a = -2$, $b = 1$ and $c = 2$.

(i) $a^3 + b^3 + c^3$

(ii) $-\frac{5}{3}a^2b + \frac{3}{4}abc + 5$

(iii) $4a^2b - \frac{3}{2}ab^2 + \frac{1}{2}abc$

(iv) $c^2 - b^3 + \frac{1}{4}a^3$

6. In the polynomials, find the degree and like terms :

(i) $5x^2 - 2x^2y + y^2 + 6yx^2$

(ii) $-3x^3 + 7x - 4x^4 - x + x^3$

(iii) $4xy + 6xy^2 + 9yx - 6$

(iv) $x^2 - 3y + 2x^2 + xy + 4$

ADDITION OF ALGEBRAIC EXPRESSIONS

Consider the following example :

Aneeta had 7 apples whereas Anil had 6 apples. To find total number of apples, we find the sum of the apples that Aneeta and Anil have together.

$$7 \text{ apples} + 6 \text{ apples} = (7 + 6) \text{ apples} = 13 \text{ apples}$$

Similarly, $9 \text{ oranges} + 8 \text{ oranges} = (9 + 8) \text{ oranges} = 17 \text{ oranges}$

We know that, an algebraic expression consists of like and unlike terms. The addition and subtraction of algebraic expressions means adding or subtracting like terms.

Example : Add $5x$ and $9x$,

$$5x + 9x = (5 + 9)x = 14x$$

(By distributive law)

Similarly, sum of $7x + 8x$ and $(-3x)$

$$\begin{aligned} 7x + 8x - 3x &= (7 + 8)x - 3x \\ &= 15x - 3x \\ &= (15 - 3)x = 12x \end{aligned}$$

Thus, we see that sum (or difference) of several like terms is another like term, whose coefficient is the sum (or difference) of the coefficients of those like terms.

Let us take some more examples :

1. Addition of positive like terms :

Example 1 : Add $7ab$, $8ab$ and $4ab$

Solution : $7ab + 8ab + 4ab = (7 + 8 + 4)ab$
 $= 19ab$

(By distributive law)

Alternative method : The sum of numerical coefficients of given like terms
 $= 7 + 8 + 4 = 19$

$\therefore$ The sum of the given like terms is another like term, whose numerical coefficient is 19.

Hence, $7ab + 8ab + 4ab = 19ab$.

Example 2 : Add $4x^2y$, $3x^2y$, $12x^2y$ and $5x^2y$.

Solution : $4x^2y + 3x^2y + 12x^2y + 5x^2y = (4 + 3 + 12 + 5)x^2y$ (By distributive law)
 $= 24x^2y$

2. Addition of negative like terms :

Example 3 : Add $-2xy$, $-5xy$ and $-7xy$

Solution : $(-2xy) + (-5xy) + (-7xy) = (-2 - 5 - 7)xy$ (By distributive law)
 $= -14xy$

Alternative method : The numerical coefficients (without negative sign) are 2, 5, 7.
 $\therefore$ Sum of numerical coefficients $= 2 + 5 + 7 = 14$.

Hence, $-2xy - 5xy - 7xy = -14xy$.

Example 4 : Add $-4x^2y^2$, $-6x^2y^2$, $-x^2y^2$ and $-5x^2y^2$.

Solution : Numerical coefficients are (without negative sign) 4, 6, 1, 5.
Sum of 4, 6, 1 and 5 $= 16$

Hence, $-4x^2y^2 - 6x^2y^2 - x^2y^2 - 5x^2y^2 = -16x^2y^2$

3. Addition of positive and negative like terms :

Example 5 : Add $4x^2y$, $6x^2y$, $-2x^2y$ and $-3x^2y$.

Solution : $4x^2y + 6x^2y - 2x^2y - 3x^2y = (4 + 6 - 2 - 3)x^2y$ (By distributive law)
 $= (10 - 5)x^2y$
 $= 5x^2y$

Example 6 : Add $14xy$, $-6xy$, $2xy$, $-7xy$ and $2xy$.

Solution : $14xy - 6xy + 2xy - 7xy + 2xy = (14xy + 2xy + 2xy) + (-6xy - 7xy)$
[Collecting positive and negative like terms together]
 $= (18xy) + (-13xy)$
 $= (18 - 13)xy$ (By distributive law)
 $= 5xy$

4. Addition of Algebraic expressions with like and unlike terms

While adding algebraic expressions having like and unlike terms, we collect different groups of like terms and find the sum. It can be done by two ways.

Horizontal method : In this method, expressions are written in a row (horizontal line), then the terms are arranged in group of like terms and then added.

Column method : In this method, each expression is written in such a way that their like terms are arranged one below the other in columns. Then terms are added columnwise.

Example 7 : Add the expressions :

(i) $2x + 3y + 4z$ and $5x + 3y + 6z$

(ii) $2x^2 + 4y^2$, $x^2 + 5y^2$ and $3x^2 + 2y^2$

Solution : (i) $(2x + 3y + 4z) + (5x + 3y + 6z) = (2x + 5x) + (3y + 3y) + (4z + 6z)$
(Collecting like terms together)
 $= 7x + 6y + 10z$

$$\begin{array}{r} \text{Alternative method : } 2x+3y+4z \\ + 5x+3y+6z \\ \hline 7x+6y+10z \end{array}$$

(Column method)

$$\begin{aligned} \text{(ii)} \quad & (2x^2 + 4y^2) + (x^2 + 5y^2) + (3x^2 + 2y^2) \\ &= (2x^2 + x^2 + 3x^2) + (4y^2 + 5y^2 + 2y^2) \quad (\text{Collecting like terms together}) \\ &= 6x^2 + 11y^2 \end{aligned}$$

$$\begin{array}{r} \text{Alternative method : } 2x^2+4y^2 \\ + 1x^2+5y^2 \\ + 3x^2+2y^2 \\ \hline 6x^2+11y^2 \end{array}$$

(Column method)

Example 8 : Add the following :

$$\text{(i)} \quad 6x^2 - 3y^2 \text{ and } 5x^2 - 2y^2 \quad \text{(ii)} \quad x - y - 2z, 4x + 6y - 2z \text{ and } -2x - 7y + 9z$$

$$\text{(iii)} \quad 4a^2 + b^2 - 3ab, 3a^2 - 7b^2 + 8ab \text{ and } a^2 - b^2 + 2ab$$

$$\begin{aligned} \text{Solution : } \text{(i)} \quad & (6x^2 - 3y^2) + (5x^2 - 2y^2) = (6x^2 + 5x^2) + (-3y^2 - 2y^2) \\ & \quad \quad \quad (\text{Collecting like terms together}) \end{aligned}$$

$$= 11x^2 + (-5y^2)$$

$$= 11x^2 - 5y^2$$

$$\begin{aligned} \text{(ii)} \quad & (x - y - 2z) + (4x + 6y - 2z) + (-2x - 7y + 9z) \\ &= (x + 4x - 2x) + (-y + 6y - 7y) + (-2z - 2z + 9z) \\ & \quad \quad \quad (\text{Collecting like terms together}) \end{aligned}$$

$$= 3x + (-2y) + 5z$$

$$= 3x - 2y + 5z$$

$$\text{Alternative method : } 1x - y - 2z$$

(Column method)

$$+ 4x + 6y - 2z$$

$$+ -2x - 7y + 9z$$

$$\hline 3x - 2y + 5z$$

$$\begin{aligned} \text{(iii)} \quad & (4a^2 + b^2 - 3ab) + (3a^2 - 7b^2 + 8ab) + (a^2 - b^2 + 2ab) \\ &= (4a^2 + 3a^2 + a^2) + (b^2 - 7b^2 - b^2) + (-3ab + 8ab + 2ab) \\ & \quad \quad \quad (\text{Collecting like terms together}) \end{aligned}$$

$$= 8a^2 + (-7b^2) + 7ab$$

$$= 8a^2 - 7b^2 + 7ab$$

SUBTRACTION OF ALGEBRAIC EXPRESSIONS

To subtract an algebraic expression from another, we change the sign (+ to - or - to +) of each term of the expression to be subtracted and then add the terms.

The difference of two like terms is a like term whose coefficient is the difference of the numerical coefficients of those like terms.

Examples : $7a^2 - 3a^2 = (7 - 3)a^2 = 4a^2$
 $5x^2 - 9x^2 = (5 - 9)x^2 = -4x^2$

Let us take some more examples :

Example 1 : Subtract :

- (i) $30xy$ from $12xy$ (ii) $2x - 3y$ from $4x - 5y$
 (iii) $a + b - 3c$ from $2c - a - 2b$

Solution : (i) $12xy - 30xy = (12 - 30)xy$ (By distributive law)
 $= -18xy$

Alternative method :

$$\begin{array}{r} 12xy \\ + 30xy \\ \hline - 18xy \end{array}$$

(ii) $(4x - 5y) - (2x - 3y) = 4x - 5y - 2x + 3y$
 $= (4x - 2x) + (3y - 5y)$ (Collecting like terms together)
 $= 2x + (-2y)$
 $= 2x - 2y$

Alternative method :

$$\begin{array}{r} 4x - 5y \\ + 2x - 3y \\ \hline 2x - 2y \end{array}$$

(iii) $(2c - a - 2b) - (a + b - 3c) = 2c - a - 2b - a - b + 3c$
 $= (2c + 3c) + (-a - a) + (-2b - b)$ (Collecting like terms together)
 $= 5c + (-2a) + (-3b)$
 $= -2a - 3b + 5c$

Alternative method :

$$\begin{array}{r} 2c - a - 2b = -a - 2b + 2c \\ - a - 2b + 2c \\ + a + b - 3c \\ \hline - 2a - 3b + 5c \end{array}$$

Example 2 : Subtract the following :

- (i) $4x^2 - 5xy + y^2$ from $14x^2 - 2xy - 3y^2$
 (ii) $a^2 - 3ab + 6b^2 - 4$ from $6ab - 5a^2 - b^2 + 7$
 (iii) $5x^3 - 2x^2 + 6xy + 4$ from $7x^3 + x^2 - xy + 9$

Solution : (i) $(14x^2 - 2xy - 3y^2) - (4x^2 - 5xy + y^2)$
 $= 14x^2 - 2xy - 3y^2 - 4x^2 + 5xy - y^2$
 $= (14x^2 - 4x^2) + (5xy - 2xy) + (-3y^2 - y^2)$
 $= 10x^2 + 3xy - 4y^2$



Puzzle

During a survey, one man asked to a boy "What's your age?" For a moment boy looked embarrassed and then boy replied, "My age three years hence multiplied by 3 and from that subtracted three times my age three years ago will give you my exact age". Then what was the age of the boy?

$$\begin{array}{r} \text{Alternative method :} \quad 14x^2 - 2xy - 3y^2 \\ - \quad 4x^2 - 5xy + y^2 \\ \hline 10x^2 + 3xy - 4y^2 \end{array}$$

$$\begin{aligned} \text{(ii)} \quad (6ab - 5a^2 - b^2 + 7) - (a^2 - 3ab + 6b^2 - 4) \\ = 6ab - 5a^2 - b^2 + 7 - a^2 + 3ab - 6b^2 + 4 \\ = (-5a^2 - a^2) + (-b^2 - 6b^2) + (6ab + 3ab) + (7 + 4) \\ = (-6a^2) + (-7b^2) + 9ab + 11 \\ = -6a^2 - 7b^2 + 9ab + 11 \end{aligned}$$

$$\begin{array}{r} \text{Alternative method :} \quad -5a^2 - b^2 + 6ab + 7 \\ \quad \quad a^2 + 6b^2 - 3ab - 4 \\ - \quad - \quad + \quad + \\ \hline -6a^2 - 7b^2 + 9ab + 11 \end{array}$$

$$\begin{aligned} \text{(iii)} \quad (7x^3 + x^2 - xy + 9) - (5x^3 - 2x^2 + 6xy + 4) \\ = 7x^3 + x^2 - xy + 9 - 5x^3 + 2x^2 - 6xy - 4 \\ = (7x^3 - 5x^3) + (x^2 + 2x^2) + (-xy - 6xy) + (9 - 4) \\ = 2x^3 + 3x^2 - 7xy + 5 \end{aligned}$$

$$\begin{array}{r} \text{Alternative method :} \quad 7x^3 + x^2 - xy + 9 \\ \quad \quad 5x^3 - 2x^2 + 6xy + 4 \\ - \quad + \quad - \quad - \\ \hline 2x^3 + 3x^2 - 7xy + 5 \end{array}$$

Example 3 : From the sum of $3x^2 - 6x + 8$ and $-4x^2 + 7x - 6$, subtract $6x^2 + 5x + 4$.

Solution : Sum of $3x^2 - 6x + 8$ and $-4x^2 + 7x - 6$

$$\begin{aligned} &= (3x^2 - 6x + 8) + (-4x^2 + 7x - 6) = (3x^2 - 4x^2) + (-6x + 7x) + (8 - 6) \\ &= -x^2 + x + 2 \end{aligned}$$

From $(-x^2 + x + 2)$, we have to subtract $(6x^2 + 5x + 4)$.

$$\begin{aligned} (-x^2 + x + 2) - (6x^2 + 5x + 4) &= -x^2 + x + 2 - 6x^2 - 5x - 4 \\ &= (-x^2 - 6x^2) + (x - 5x) + (2 - 4) \\ &= -7x^2 - 4x - 2 \end{aligned}$$

Example 4 : What must be added to $7x^3 - 3x^2 + 6x + 4$ to make the sum $x^3 - 5x^2 - x + 1$?

Solution : Let X must be added, so

$$(7x^3 - 3x^2 + 6x + 4) + X = x^3 - 5x^2 - x + 1$$

$$\Rightarrow X = (x^3 - 5x^2 - x + 1) - (7x^3 - 3x^2 + 6x + 4)$$

$$= x^3 - 5x^2 - x + 1 - 7x^3 + 3x^2 - 6x - 4$$

$$= (x^3 - 7x^3) + (-5x^2 + 3x^2) + (-x - 6x) + (1 - 4)$$

(Collecting like terms together)

$$= -6x^3 - 2x^2 - 7x - 3$$

$\therefore (-6x^3 - 2x^2 - 7x - 3)$ must be added.

Example 5 : What should be subtracted from $p^3 - 7q^2 + 5p - 7$ to get $q^2 - 2p - 6$?

Solution : Let X to be subtracted, then

$$(p^3 - 7q^2 + 5p - 7) - X = q^2 - 2p - 6$$

or

$$X = (p^3 - 7q^2 + 5p - 7) - (q^2 - 2p - 6)$$

$$= p^3 - 7q^2 + 5p - 7 - q^2 + 2p + 6$$

$$= p^3 + (-7q^2 - q^2) + (5p + 2p) + (-7 + 6)$$

(Collecting like terms together)

$$= p^3 + (-8q^2) + 7p + (-1)$$

$$= p^3 - 8q^2 + 7p - 1$$

Hence, $p^3 - 8q^2 + 7p - 1$ should be subtracted.

Example 6 : How much is $x + 3y - 4z$ greater than $3x - 2y + z$?

Solution : Required expression = $(x + 3y - 4z) - (3x - 2y + z)$

$$= x + 3y - 4z - 3x + 2y - z$$

$$= (x - 3x) + (3y + 2y) + (-4z - z)$$

(Collecting like terms together)

$$= -2x + 5y - 5z$$

Example 7 : How much does $-2a^2 + 6b + 9$ exceed $4a^2 - 2b + 7$?

Solution : Required expression = $(-2a^2 + 6b + 9) - (4a^2 - 2b + 7)$

$$= -2a^2 + 6b + 9 - 4a^2 + 2b - 7$$

$$= (-2a^2 - 4a^2) + (6b + 2b) + (9 - 7)$$

(Collecting like terms together)

$$= -6a^2 + 8b + 2$$

Example 8 : Simplify :

$$(i) (9x^3 - 7x^2 - 5) - (-2x^3 + 4x^2 + 1) + (6x^3 - 5)$$

$$(ii) (8a^3b - 7a^2b^2 - 4ab) - (5a^3b + 8a^2b^2 - 7ab) + (a^3b + ab)$$

Solution : We have,

$$(i) (9x^3 - 7x^2 - 5) - (-2x^3 + 4x^2 + 1) + (6x^3 - 5)$$

$$= 9x^3 - 7x^2 - 5 + 2x^3 - 4x^2 - 1 + 6x^3 - 5$$

$$= (9x^3 + 2x^3 + 6x^3) + (-7x^2 - 4x^2) + (-5 - 1 - 5)$$

(Collecting like terms together)

$$= 17x^3 + (-11x^2) + (-11)$$

$$= 17x^3 - 11x^2 - 11$$

$$(ii) (8a^3b - 7a^2b^2 - 4ab) - (5a^3b + 8a^2b^2 - 7ab) + (a^3b + ab)$$

$$= 8a^3b - 7a^2b^2 - 4ab - 5a^3b - 8a^2b^2 + 7ab + a^3b + ab$$

$$= (8a^3b - 5a^3b + a^3b) + (-7a^2b^2 - 8a^2b^2) + (-4ab + 7ab + ab)$$

(Collecting like terms together)

$$= 4a^3b + (-15a^2b^2) + (4ab)$$

$$= 4a^3b - 15a^2b^2 + 4ab$$



EXERCISE 7.3

1. Add the following :

$$(i) x^2y, 4x^2y, -2x^2y$$

$$(ii) 6a^2, -4a^2, a^2, -2a^2$$

$$(iii) 2x, 3x, -x, +y$$

$$(iv) 4x^2, 3x^2, -2, -5x^2$$

2. Simplify each of the following :

$$(i) 7x + 9x$$

$$(ii) 3x^2 + 8x^2$$

$$(iii) -2x^2y + x^2y$$

$$(iv) 9a^2 - 4a^2$$

$$(v) 4x^2y^2 + 8x^2y^2$$

$$(vi) 4ax - 9ax$$

3. Add the following :

$$(i) 2xy, -5xy, 6xy$$

$$(ii) 3a^2b, -9a^2b, 6a^2b$$

$$(iii) 10a^3b, -6a^3b, 2a^3b$$

$$(iv) 7x^2y^2, 8x^2y^2, -9x^2y^2$$

4. Find the sum of the polynomials :

$$(i) 7x - 4y, 2x + 6y$$

$$(ii) 9x^2 + 5x, 6x^2 - 2x$$

$$(iii) 6a^2b^2 - 4a, -7a^2b^2 - 2a$$

$$(iv) 11ab - 7a^2, -16ab + 11a^2$$

$$(v) 18xy - 7, -11xy + 11$$

$$(vi) 2a + b, 6a^2 - 7a - b$$

$$(vii) 4a^3b - 7a^2b^2 + 8ab, -6a^3b + 6a^2b^2 - ab$$

$$(viii) 12x^2 - x^3 + 7x + 6, 2x^3 + 6x^2 - 6x - 6$$

5. Add the following polynomials :

$$(i) 2a + b + c, a - b - c, 3a + b$$

$$(ii) 6x^2 + 5x + 2, x^2 - 5, x^2 - 4x + 1$$

$$(iii) x - 3y + z, -3x + 2y - z, 6y + 4z$$

$$(iv) 7a + 2b - 6ab, 4ab + 3b, -6a + 2b + 2ab$$

6. Subtract :

$$(i) 4a \text{ from } 7a$$

$$(ii) 8a^2b^2 \text{ from } a^2b^2$$

$$(iii) 3a \text{ from } 2b$$

$$(iv) -3x \text{ from } -9y$$

7. Subtract the following polynomial :

(i) $4x^3 - 5x^2 + 6$ from $3x^3 - 7x^2 - 5$ (ii) $-x^2 - 4z$ from $2x^2 + 3y - 7z$

(iii) $2x^2y^2 - 5xy + y^2$ from $2y^2 + 6x^2y^2 - xy$

8. From

(i) $2x - 4y + 6$ take away $3y - 11$ (ii) $3 + a + a^2$ take away $-5 + 4a^2$

(iii) $x^2 - 2x + 1$ take away $4x - 5$ (iv) $3x^3 + 2xy - x^2$ take away $7x^3 - xy + 4x^2$

9. From the sum of $5a^2 - 3a + 2$ and $-8a^2 + 5a + 6$, subtract $a^2 + 7a - 4$.

10. What must be added to $2x^2y^2 - 5x + 7$ to get the sum $-5x^2y^2 + 7x - 9$?

11. How much is $2x + 3y - 8z$ greater than $-2x + 7y + 9z$?

12. What must be subtracted from $a^3b + 6a^2b^2 - 7ab + 4$ to get $-7a^2b^2 + 2a^3b + 6ab - 1$?

13. If $A = x^2 + xy - 6$, $B = 6xy - 2x^2 + 1$ and $C = 3x^2 + 7 - 3xy$, find

(i) $A + B + C$

(ii) $A - B + C$

(iii) $A + B - C$

14. Subtract $16x - 11y + 2z$ from the sum of $7x + 6y + z$ and $-9x + 3y - 2z$.

15. How much does $x^2 - 2ax + 5$ exceed $3x^2 - 5x + 6$?

16. If $P = 3x^2 + 7x + 8$, $Q = 2x^2 + x - 9$ and $R = -5x^2 - 8x + 1$, show that $P + Q + R = 0$.

17. Simplify :

(i) $(2x^3 - 3x^2 - 8) - (-5x^3 - 2x^2 + 7) + (x^2 - 3)$

(ii) $(2 + a - a^2) - (6 - 3a + 4a^2) - (3 + 5a + a^2)$



FACTS TO REMEMBER

- The letters which are used to represent numbers are called literals.
- The literal, numbers as well as their combinations follow all the rules (on signs) of addition, subtraction, multiplication and division of numbers along with the properties of these operations.
- A symbol having a fixed numerical value is called a constant and a symbol which takes various numerical values is called a variable.
- A combination of variables and constants connected by the signs of fundamental operations of $+$, $-$, $\times$ and $\div$ is called an algebraic expression.
- Expressions are made up of terms. Terms are added to make an expression. For example, $2x$ and -7 gives the expression $2x - 7$.
- Any expression with one or more terms is called a polynomial. An expression having one term is called a monomial, having two terms is called a binomial and having three terms is called a trinomial.

- Terms having the same algebraic factors are called like terms. Terms which have different algebraic factors are called unlike terms.
- The sum or difference of several like terms is another like term, whose coefficient is the sum or difference of those like terms.
- In adding or subtracting algebraic expressions, we find out sum or difference of like terms in each group.
- In subtraction we change the sign (+ to - or - to +) of each term of the expression to be subtracted and then we add the two expressions.
- The exponent of the term containing the highest power of the variable in a polynomial is called the degree of the polynomial.
- Linear polynomial is a polynomial of degree 1, quadratic polynomial is a polynomial of degree 2 and cubic polynomial is a polynomial of degree 3.
- Degree of a non-zero constant polynomial is zero.
- Addition and subtraction operation cannot be performed on unlike terms.



MULTIPLE CHOICE QUESTIONS

Choose the correct alternatives in each of the following :

- The degree of the polynomial $a^2b + 6ab^2 + 5a^2b^3$ is
 (a) 3 (b) 2 (c) 5 (d) 1
- The value of expression $\frac{3x^2-5}{2-x}$ for $x = -2$ is
 (a) $\frac{7}{4}$ (b) 0 (c) $-\frac{7}{4}$ (d) infinite
- What must be added to $1 - 2x$ to get $-2 + 5x$?
 (a) $3 - 7x$ (b) $-1 + 3x$ (c) $7x - 3$ (d) $3x + 1$
- The statement, "4 times a number p is added to 3 times a number q " is expressed in algebraic expression as
 (a) $(4 + p) + (3 + q)$ (b) $4p + 3q$ (c) $3p + 4q$ (d) None of these
- The expression $2z - 5x + 7y$ is a
 (a) binomial (b) trinomial
 (c) monomial (d) cubic polynomial

6. Coefficient of $-x^2$ in $-5x^3$ is
 (a) -5 (b) $5x$ (c) $-5x$ (d) $-5x^2$
7. $7x^2y$ is a
 (a) cubic polynomial (b) quadratic polynomial
 (c) linear polynomial (d) none of these
8. On subtracting $x^2 - y^2$ from $y^2 - x^2$, we get
 (a) 0 (b) $2y^2$ (c) $-2(x^2 - y^2)$ (d) $-2x^2$
9. What must be subtracted from $3x^2y$ to get $-7x^2y$?
 (a) $4x^2y$ (b) $10x^2y$ (c) $-4x^2y$ (d) $-10x^2y$
10. If $p = -10$, then the value of $p^2 - 2p - 100$ is
 (a) 20 (b) -220 (c) 180 (d) -20



MENTAL MATHS CORNER

Fill in the blanks :

1. A symbol which takes various numerical values is called a
2. The terms not having same variable factors are called
3. The degree of non-zero constant polynomial is
4. Linear polynomial is a polynomial of degree
5. A polynomial of degree three is called a
6. A binomial has terms.
7. The value of $\frac{5-x}{x}$, when $x = -2$ is
8. On subtracting $3xy$ from $-7xy$, we get
9. must be added to $a^3 - b^3$ to get $-2b^3$.
10. must be subtracted from $-10ab^2c$ to get $4ab^2c$.



REVIEW EXERCISE

- Write the degree of the following polynomials :
(i) $6x^2y - 2x^2y^3 - xy + 7$ (ii) $xyz - x^3 + 6x^2$
- For the polynomial $9y^2 - 4x + 7y^3 - 5$, find
(i) its degree (ii) number of terms
(iii) constant term (iv) coefficient of x .
- Simplify : $5x^2 + 7 - 2x + 3x^2 + 4x + 1$
- Find the value of the expression :
(i) $-\frac{2}{3}x^2 + 5x - 2$ for $x = -3$ (ii) $\frac{3a+2b-5}{a+b}$ for $a=2, b=-1$
- Add :
(i) $7x^3 + 2x^2 - 5x - 7$ and $-5x^2 + x^3 + 4x - 5$ (ii) $3x - x^2 + 1$, $2x$ and $1 + x^2$
- Subtract :
(i) $-10x + 8y$ from $2y - 5x$ (ii) $9a^3 - 7a^2 + 5a + 7$ from $-4a^3 + 5a^2 - 3a - 2$
- What should be added to $7x^2 - 5x + 1$ to get $-3x^2 + 4x + 7$?
- Subtract $2x + 6$ from the sum of $x^2 + 4x + 6$ and $8 - x^2 + 6x$.
- What should be subtracted from $17x^3 - 8x^2 + 11$ to obtain $14x^3 + 8x^2 - 5x + 6$?
- If $P = x^2 + 6$, $Q = 3x^2 - x + 2$ and $R = x^2 - 4x$, then find $P + Q - R$.
- Evaluate $7x - 2y + 5xy + \frac{1}{3}x^2$ for $x = -1$ and $y = 2$.



HOTS QUESTIONS

- If $x = 3$, then find the value of k :
 $2(kx + 5) = 7 + 3x$.
- Change the following statement into algebraic expression.
'The subtraction of $x - 2$ from y is added to $4x$ and its difference from x is subtracted from y .'
- Evaluate the expression : $\frac{(x+y)(x^2 - xy + y^2)}{x^3 + y^3}$ for $x = 1$ and $y = -2$.



ANSWERS

Exercise 7.1

- (i) $5x^4, -3x^3, 2x, 7$ (ii) $a^2, b^2, 2ab$
 (iii) $a^3, -ab, b^2$ (iv) $12x^2, 5x, -8$
 (v) $3x^3, -7x^2y^2, 7$ (vi) $4x^2y^3, -7xy, 4$
- (i) -8 (ii) $\frac{7}{9}$ (iii) $\frac{1}{2}$ (iv) -3
- (i) binomial (ii) monomial
 (iii) trinomial (iv) monomial
 (v) monomial (vi) binomial
 (vii) quadrinomial (viii) monomial
 (ix) trinomial (x) binomial
- (i) -2 (ii) $15y^3$ (iii) $-\frac{7}{3}yz^2$ (iv) $-\frac{3}{2}$
 (v) $-\frac{7}{2}a$ (vi) $-\frac{5}{9}y$ (vii) $4y^2$ (viii) $-6y^3$
- (i) Difference of x and y (ii) 4 times sum of x and y
 (iii) 5 added to seven times y
 (iv) b subtracted from twice of a
 (v) 6 subtracted from quotient of x by y
 (vi) x multiplied by itself
 (vii) 7 added to the 3 times the product of m and n
 (viii) sum of a and b subtracted from the product of a and b .
- (i) unlike terms (ii) like terms
 (iii) unlike terms (iv) unlike terms
 (v) like terms (vi) like terms
- (i) $3x^2, -x^2$ (ii) $bc, -4bc$ (iii) $-4x^2y, -6yx^2$
 (iv) $x^2yz, zx^2y; yxz^2, 6xyx^2$
- (i) $x^2 + y^2$ (ii) $9 - xy$ (iii) $xy^2 - 5$
 (iv) $2xy + 8$ (v) $(x - 2y) + 6$
 (vi) $x^2 + 7$

Exercise 7.2

- (i) 4 (ii) 3 (iii) 2 (iv) 4
 (v) 4 (vi) 3
- (i) monomial, quadratic polynomial
 (ii) binomial, linear polynomial
 (iii) binomial, linear polynomial
 (iv) trinomial quadratic polynomial
 (v) trinomial, cubic polynomial
 (vi) monomial, cubic polynomial.
- (i) linear (ii) quadratic
 (iii) cubic (iv) cubic
 (v) linear (vi) cubic
- (i) -2 (ii) 0 (iii) $\frac{1}{3}$ (iv) 7 (v) -7
- (i) 1 (ii) $-\frac{14}{3}$ (iii) 17 (iv) 1
- (i) 3, $(-2x^2y, 6yx^2)$ (ii) 4, $(-3x^3, x^3), (7x, -x)$
 (iii) 3, $(4xy, 9yx)$ (iv) 2, $(x^2, 2x^2)$

Exercise 7.3

- (i) $3x^2y$ (ii) a^2 (iii) $4x + y$ (iv) $2x^2 - 2$
- (i) $16x$ (ii) $11x^2$ (iii) $-x^2y$ (iv) $5a^2$
 (v) $12x^2y^2$ (vi) $-5ax$
- (i) $3xy$ (ii) 0 (iii) $6a^3b$ (iv) $6x^2y^2$
- (i) $9x + 2y$ (ii) $15x^2 + 3x$ (iii) $-a^2b^2 - 6a$
 (iv) $-5ab + 4a^2$ (v) $7xy + 4$ (vi) $6a^2 - 5a$
 (vii) $-2a^3b - a^2b^2 + 7ab$ (viii) $x^3 + 18x^2 + x$
- (i) $6a + b$ (ii) $8x^2 + x - 2$ (iii) $-2x + 5y + 4z$
 (iv) $a + 7b$
- (i) $3a$ (ii) $-7a^2b^3$ (iii) $2b - 3a$ (iv) $3x - 9y$
- (i) $-x^3 - 2x^2 - 11$ (ii) $3x^2 + 3y - 3z$
 (iii) $y^2 + 4x^2y^2 + 4xy$
- (i) $2x - 7y + 17$ (ii) $-3a^2 + a + 8$
 (iii) $x^3 - 6x + 6$ (iv) $-4x^3 + 3xy - 5x^2$
- $-4a^2 - 5a + 12$

$$10. -7x^2y^2 + 12x - 16$$

$$11. 4x - 4y - 17z$$

$$12. -a^3b + 13a^2b^2 - 13ab + 5$$

$$13. (i) 2x^2 + 4xy + 2 \quad (ii) 6x^2 - 8xy$$

$$(iii) -4x^2 + 10xy - 12$$

$$14. -18x + 20y - 3z$$

$$15. -2x^2 - 2ax + 5x - 1$$

$$17. (i) 7x^3 - 18 \quad (ii) -6a^2 - a - 7$$

Multiple Choice Questions

- | | | | |
|--------|---------|--------|--------|
| 1. (c) | 2. (a) | 3. (c) | 4. (b) |
| 5. (b) | 6. (b) | 7. (a) | 8. (c) |
| 9. (b) | 10. (a) | | |

Mental Maths Corner

- | | | |
|-------------|---------------------|---------|
| 1. variable | 2. unlike terms | 3. zero |
| 4. one | 5. cubic polynomial | 6. two |

$$7. -\frac{7}{2}$$

$$8. -10xy$$

$$9. -a^3 - b^3$$

$$10. -14ab^2c$$

Review Exercise

$$1. (i) 5 \quad (ii) 3 \quad 2. (i) 3 \quad (ii) 4 \quad (iii) -5 \quad (iv) -4$$

$$3. 8x^2 + 2x + 8 \quad 4. (i) -23 \quad (ii) -1$$

$$5. (i) 8x^3 - 3x^2 - x - 12 \quad (ii) 5x + 2$$

$$6. (i) 5x - 6y \quad (ii) -13a^2 + 12a^2 - 8a - 9$$

$$7. -10x^2 + 9x + 6$$

$$8. 8x + 8$$

$$9. 3x^3 - 16x^2 + 5x + 5$$

$$10. 3x^2 + 3x + 8$$

$$11. -\frac{62}{3}$$

HOTS Questions

$$1. k = 1 \quad 2. y - [x - (4x + y - (x - 2))]$$

$$3. 1$$